

CS 170 Lab 5: Strings

For each program below, give what is printed. Each prints exactly 10 lines. In the skills assessment, you will need to be able to trace code without being able to execute it, so practice here tracing without running the code. You may however check your work by executing the code if you wish.

BrushPencilMarker

```
public static void main(String[] args) {
    String brush = "pencil";
    String pencil = "marker";
    int marker = 3;
    System.out.println(pencil.charAt(1));
    System.out.println("pencil".charAt(1));
    System.out.println(brush + pencil);
    System.out.println(marker + pencil);
    System.out.println(pencil.charAt(marker));
    System.out.println(brush.substring(1, marker));
    System.out.println(("marker" + pencil).substring(4, 7));
    System.out.println(("" + pencil.charAt(2) +
        brush.substring(marker)).substring(1, 3));
    System.out.println("brush" + brush.length());
    System.out.println(pencil.substring(0, marker) + "*" +
        pencil.substring(marker));
}
```

Handwritten annotations for tracing:

- `marker`: m a r k e r (indices 0-6)
- `pencil`: p e n c i l (indices 0-5)
- `brush`: b r u s h (indices 0-5)
- `marker` variable: 3
- Line 1: `pencil.charAt(1)` → 'e'
- Line 2: `"pencil".charAt(1)` → 'e'
- Line 3: `brush + pencil` → "pencilmarker"
- Line 4: `marker + pencil` → "3marker"
- Line 5: `pencil.charAt(marker)` → 'm' (index 3)
- Line 6: `brush.substring(1, marker)` → "en" (indices 1-2)
- Line 7: `("marker" + pencil).substring(4, 7)` → "erm" (indices 4-6 of "markerpencil")
- Line 8: `("" + pencil.charAt(2) + brush.substring(marker)).substring(1, 3)` → "ci" (pencil[2]='n', brush[3]='s', then substring(1,3) of "ns" is "ci")
- Line 9: `"brush" + brush.length()` → "brush 6"
- Line 10: `pencil.substring(0, marker) + "*" + pencil.substring(marker)` → "mar*ker"

Answer:

1. "e"
2. "e"
3. "pencilmarker"
4. "3marker"
5. "m"
6. "en"
7. "erm"
8. "ci"
9. "brush 6"
10. "mar*ker"

Hey

```

public static void main(String[] args) {
    String x = "hey"; // length = 3
    int n = x.length(); // = 3
    String y = "";
    int i = 0; // while i < 3
    while (i < n) { // 0 < 3, 1 < 3, 2 < 3, 3 < 3
        y += x.charAt(n - (i + 1)); // y = "y", y = "ye", y = "yeh"
        System.out.println(i); // 0, 1, 2
        System.out.println("*" + x.substring(i)); // "*hey", "*ey", "*y"
        System.out.println(y); // "y", "ye", "yeh"
        i++; // i = 1, i = 2, i = 3 stop
    }
    System.out.println(x.charAt(x.length() - 1) +
        x.charAt(3 - 1) + x.substring(1, 2) + "a" +
        x.charAt(0)); // y + e + a + y
}

```

Loop 1: i = 0
 y += x.charAt(3 - (0 + 1)) → y += x.charAt(2) →
 y += 'y'
 x.substring(i) = x.substring(0) =
 "hey"

Loop 2: i = 1, y = "y"
 y += x.charAt(3 - (1 + 1)) → charAt(1)
 y += 'e'
 x.substring(i) = x.substring(1) =
 "ey"

Loop 3: i = 2, y = "ye"
 y += x.charAt(3 - (2 + 1)) → charAt(0)
 y += 'h'
 x.substring(i) = x.substring(2) =
 "h"

Answer:

1. "0"

2. "*hey"

3. "y"

4. "1"

5. "*ey"

6. "ye"

7. "2"

8. "*y"

9. "yeh"

10. "Yeah"